

(TR-104) PYTHON WITH AI/ML

Training Day 58 Report

Date: 1 April 2026

Topic Covered: AI Agents with MCP, Tool Calling & System Debugging

The fifty-eighth day of training focused on building AI agents using MCP (Model Context Protocol), integrating tool calling with LLMs, and understanding real-world limitations of models in agentic workflows. The session also included testing MCP systems, debugging tool behavior, and analyzing frontend validation issues in a web application.

- Learned how AI agents interact with external tools using MCP (Model Context Protocol).
- Understood how tool calling enables LLMs to perform actions instead of only generating text responses.
- Built and tested an MCP-based tool system using FastMCP.
- Created multiple tools including `get_current_time()`, `add_numbers(a, b)`, and `reverse_text(text)`.
- Connected MCP server with LangChain agent using `MultiServerMCPClient`.
- Understood dynamic tool selection based on user queries in agent workflows.
- Experimented with async execution of agents using `.ainvoke()` for non-blocking calls.
- Studied MCP transport mechanisms including stdio and streamable HTTP communication layers.
- Tested tool calling behavior with different LLMs and observed limitations in llama3 for structured tool execution.
- Switched to minstral-3:3b model for more stable and reliable tool calling performance.
- Understood the importance of model compatibility when building agent-based systems.
- Learned about uv package manager for faster dependency management and project setup compared to pip.

- Performed website testing and identified multiple frontend validation and usability issues.
- Found missing input validation for phone numbers, guest count, and event date constraints.
- Identified CORS-related issue in form submission causing failure responses.
- Noted missing flexibility in “Other” event type input handling.
- Strengthened understanding of real-world debugging in full-stack AI applications combining frontend, backend, and LLM systems.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/agents_mcp
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/building-an-intelligent-ai-agent-using-mcp-langchain-and-ollama-1c646fae9e05>